Task

Task: Install Prometheus and Grafana on a Linux EC2 machine, connect Prometheus to Grafana, and create a dashboard to view metrics.

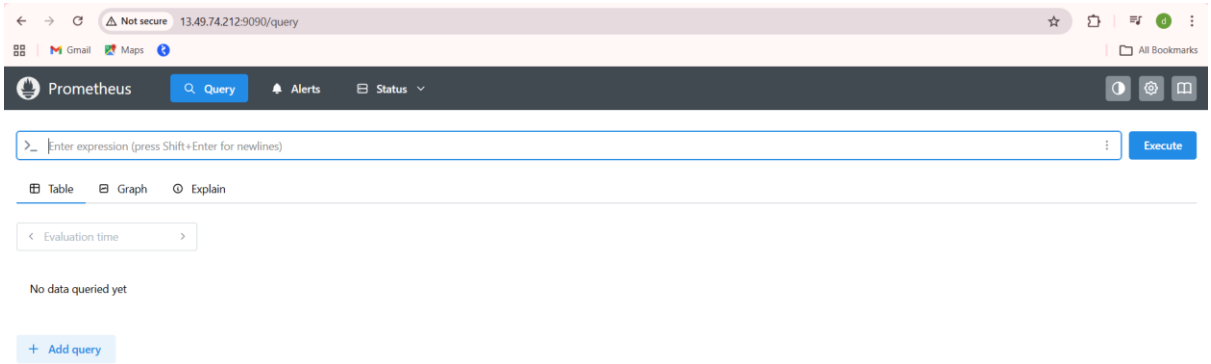
1. **Install Prometheus**

```
ubuntu@ip-172-31-24-230:~$ wget https://github.com/prometheus/prometheus/release
s/download/v3.8.1/prometheus-3.8.1.linux-amd64.tar.gz
--2026-03-04 06:38:48--  https://github.com/prometheus/prometheus/releases/downl
oad/v3.8.1/prometheus-3.8.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 4.225.11.194
Connecting to github.com (github.com)|4.225.11.194|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release
-asset/6838921/320d0920-0d04-4920-8106-701f5db9519a?sp=r&sv=2018-11-09&sr=b&spr=
https&se=2026-03-04T07%3A19%3A46Z&rscd=attachment%3B+filename%3Dprometheus-3.8.1
.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aed
d-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-03-04T06%3A18
%3A53Z&ske=2026-03-04T07%3A19%3A46Z&sks=b&skv=2018-11-09&sig=Lx9YG0ROijy1bslDi8k
byC9IDdB%2BzVcq9BMYi2s%2F%2BqU%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc
3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tI
iwia2V5Ijoia2V5MSIsImV4cCI6MTc3MjYwOTkyOCwibmJmIjoxNzcyNjA2MzI4LCJwYXRoIjoicmVsZ
WFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.M1ZK6dtIfsJKO-7ffjqPQl
wOOLFDRxByXUPrxyeAF2Y&response-content-disposition=attachment%3B%20filename%3Dpr
ometheus-3.8.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stre
am [following]
--2026-03-04 06:38:48--  https://release-assets.githubusercontent.com/github-pro
duction-release-asset/6838921/320d0920-0d04-4920-8106-701f5db9519a?sp=r&sv=2018-
11-09&sr=b&spr=https&se=2026-03-04T07%3A19%3A46Z&rscd=attachment%3B+filename%3Dp
rometheus-3.8.1.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d41
0-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=202
6-03-04T06%3A18%3A53Z&ske=2026-03-04T07%3A19%3A46Z&sks=b&skv=2018-11-09&sig=Lx9Y
G0ROijy1bslDi8kbyC9IDdB%2BzVcq9BMYi2s%2F%2BqU%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJ
IUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcm
NvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MjYwOTkyOCwibmJmIjoxNzcyNjA2MzI4LC
JwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.M1ZK6dt
IfsJKO-7ffjqPQlwOOLFDRxByXUPrxyeAF2Y&response-content-disposition=attachment%3B%
20filename%3Dprometheus-3.8.1.linux-amd64.tar.gz&response-content-type=applicati
on%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent
.com)... 185.199.108.133, 185.199.110.133, 185.199.109.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercon
tent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 133880557 (128M) [application/octet-stream]
Saving to: ‘prometheus-3.8.1.linux-amd64.tar.gz’

prometheus-3.8.1.li 100%[===================>] 127.68M  28.2MB/s    in 4.6s

2026-03-04 06:38:53 (28.0 MB/s) - ‘prometheus-3.8.1.linux-amd64.tar.gz’ saved [1
33880557/133880557]

ubuntu@ip-172-31-24-230:~$ tar -xvzf prometheus-3.8.1.linux-amd64.tar.gz
prometheus-3.8.1.linux-amd64/
prometheus-3.8.1.linux-amd64/NOTICE
prometheus-3.8.1.linux-amd64/LICENSE
prometheus-3.8.1.linux-amd64/prometheus.yml
prometheus-3.8.1.linux-amd64/promtool
prometheus-3.8.1.linux-amd64/prometheus
```

```
ubuntu@ip-172-31-24-230:~$ tar -xvf prometheus-*.linux-amd64.tar.gz
prometheus-3.8.1.linux-amd64/
prometheus-3.8.1.linux-amd64/NOTICE
prometheus-3.8.1.linux-amd64/LICENSE
prometheus-3.8.1.linux-amd64/prometheus.yml
prometheus-3.8.1.linux-amd64/promtool
prometheus-3.8.1.linux-amd64/prometheus
ubuntu@ip-172-31-24-230:~$ ls
prometheus-3.8.1.linux-amd64  prometheus-3.8.1.linux-amd64.tar.gz
ubuntu@ip-172-31-24-230:~$ cd prometheus-*.linux-amd64
ubuntu@ip-172-31-24-230:~/prometheus-3.8.1.linux-amd64$ ls
LICENSE  NOTICE  prometheus  prometheus.yml  promtool
ubuntu@ip-172-31-24-230:~/prometheus-3.8.1.linux-amd64$ ./prometheus --config.fi
le=prometheus.yml
time=2026-03-04T06:42:55.748Z level=INFO source=main.go:1557 msg="updated GOGC"
old=100 new=75
time=2026-03-04T06:42:55.749Z level=INFO source=main.go:688 msg="Leaving GOMAXPR
OCS=2: CPU quota undefined" component=automaxprocs
time=2026-03-04T06:42:55.749Z level=INFO source=memlimit.go:198 msg="GOMEMLIMIT
is updated" component=automemlimit package=github.com/KimMachineGun/automemlimit
/memlimit GOMEMLIMIT=862757683 previous=9223372036854775807
time=2026-03-04T06:42:55.749Z level=INFO source=main.go:730 msg="No time or size
 retention was set so using the default time retention" duration=15d
time=2026-03-04T06:42:55.749Z level=INFO source=main.go:781 msg="Starting Promet
heus Server" mode=server version="(version=3.8.1, branch=HEAD, revision=ed753444
ffec98097399d0cfa9073c70a840b812)"
time=2026-03-04T06:42:55.749Z level=INFO source=main.go:786 msg="operational inf
ormation" build_context="(go=go1.25.5, platform=linux/amd64, user=root@fcf67c824
170, date=20251216-08:46:44, tags=netgo,builtinassets)" host_details="(Linux 6.1
4.0-1018-aws #18~24.04.1-Ubuntu SMP Mon Nov 24 19:46:27 UTC 2025 x86_64 ip-172-3
1-24-230 (none))" fd_limits="(soft=1048575, hard=1048576)" vm_limits="(soft=unli
mited, hard=unlimited)"
time=2026-03-04T06:42:55.754Z level=INFO source=web.go:663 msg="Start listening
for connections" component=web address=0.0.0.0:9090
time=2026-03-04T06:42:55.754Z level=INFO source=main.go:1301 msg="Starting TSDB
..."
time=2026-03-04T06:42:55.756Z level=INFO source=tls_config.go:354 msg="Listening
 on" component=web address=[::]:9090
time=2026-03-04T06:42:55.756Z level=INFO source=tls_config.go:357 msg="TLS is di
sabled." component=web http2=false address=[::]:9090
time=2026-03-04T06:42:55.762Z level=INFO source=head.go:666 msg="Replaying on-di
sk memory mappable chunks if any" component=tsdb
time=2026-03-04T06:42:55.762Z level=INFO source=head.go:752 msg="On-disk memory
mappable chunks replay completed" component=tsdb duration=6.858µs
time=2026-03-04T06:42:55.762Z level=INFO source=head.go:760 msg="Replaying WAL,
this may take a while" component=tsdb
time=2026-03-04T06:42:55.763Z level=INFO source=head.go:833 msg="WAL segment loa
ded" component=tsdb segment=0 maxSegment=0 duration=196.789µs
time=2026-03-04T06:42:55.763Z level=INFO source=head.go:870 msg="WAL replay comp
leted" component=tsdb checkpoint_replay_duration=36.018µs wal_replay_duration=22
9.114µs wbl_replay_duration=150ns chunk_snapshot_load_duration=0s mmap_chunk_rep
lay_duration=6.858µs total_replay_duration=313.226µs
time=2026-03-04T06:42:55.767Z level=INFO source=main.go:1322 msg="filesystem inf
ormation" fs_type=EXT4_SUPER_MAGIC
time=2026-03-04T06:42:55.767Z level=INFO source=main.go:1325 msg="TSDB started"
time=2026-03-04T06:42:55.767Z level=INFO source=main.go:1510 msg="Loading config
uration file" filename=prometheus.yml
time=2026-03-04T06:42:55.770Z level=INFO source=main.go:1550 msg="Completed load
```

```
e manager..." component="rule manager"
time=2026-03-04T06:42:55.770Z level=INFO source=main.go:1286 msg="Server is read
y to receive web requests."
```

2. open in browser

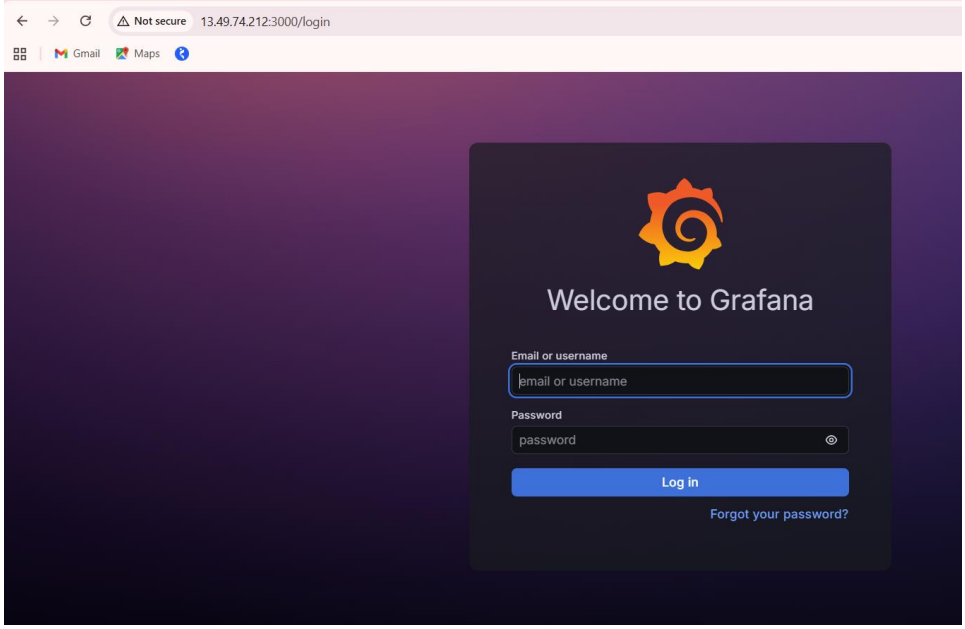


3. Install Grafana

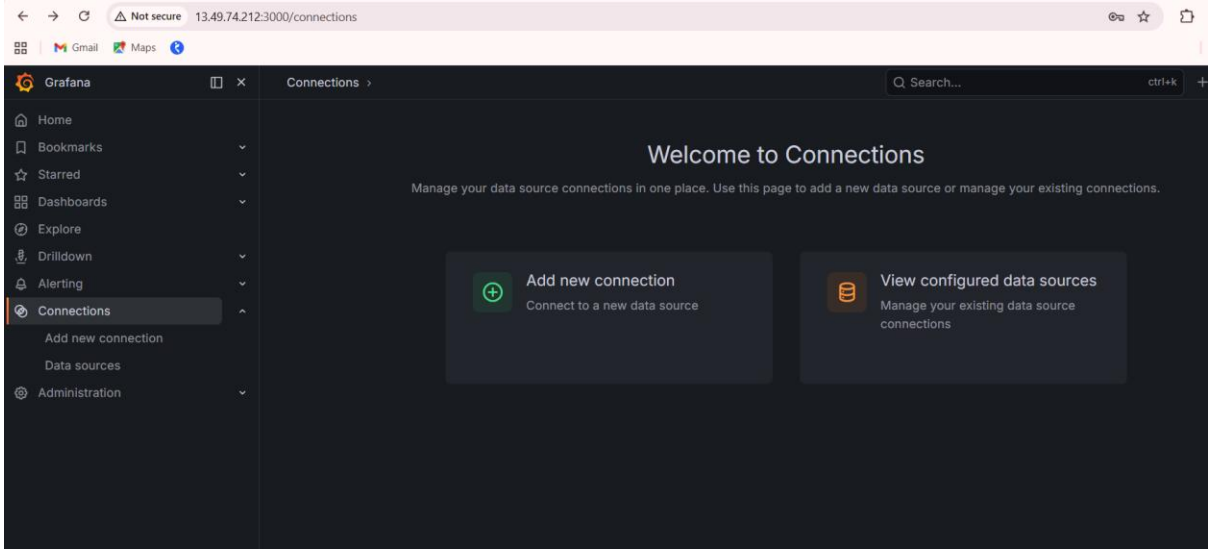
```
ubuntu@ip-172-31-24-230:~$ sudo apt-get install -y software-properties-common
sudo apt-get install -y apt-transport-https
sudo mkdir -p /etc/apt/keyrings/

wget -q -O - https://packages.grafana.com/gpg.key | sudo gpg --dearmor -o /etc/apt/keyrings/grafana.gpg

echo "deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://packages.grafana.com/oss/deb stable main" | su
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
software-properties-common is already the newest version (0.99.49.4).
software-properties-common set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 99 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apt-transport-https is already the newest version (2.8.3).
0 upgraded, 0 newly installed, 0 to remove and 99 not upgraded.
File '/etc/apt/keyrings/grafana.gpg' exists. Overwrite? (y/N) y
deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://packages.grafana.com/oss/deb stable main
ubuntu@ip-172-31-24-230:~$ sudo apt update
sudo apt install grafana -y
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 https://packages.grafana.com/oss/deb stable InRelease [7660 B]
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 https://packages.grafana.com/oss/deb stable/main amd64 Packages [472 kB]
Fetched 480 kB in 2s (260 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
99 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  musl
The following NEW packages will be installed:
  grafana musl
0 upgraded, 2 newly installed, 0 to remove and 99 not upgraded.
Need to get 197 MB of archives.
After this operation, 725 MB of additional disk space will be used.
Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 musl amd64 1.2.4-2 [416 kB]
Get:2 https://packages.grafana.com/oss/deb stable/main amd64 grafana amd64 12.4.0 [197 MB]
Fetched 197 MB in 9s (22.5 MB/s)
Selecting previously unselected package musl:amd64.
(Reading database ... 71756 files and directories currently installed.)
Preparing to unpack .../musl_1.2.4-2_amd64.deb ...
Unpacking musl:amd64 (1.2.4-2) ...
Selecting previously unselected package grafana.
Preparing to unpack .../grafana_12.4.0_amd64.deb ...
Unpacking grafana (12.4.0) ...
Setting up musl:amd64 (1.2.4-2) ...
Setting up grafana (12.4.0) ...
info: Selecting UID from range 100 to 999 ...
```

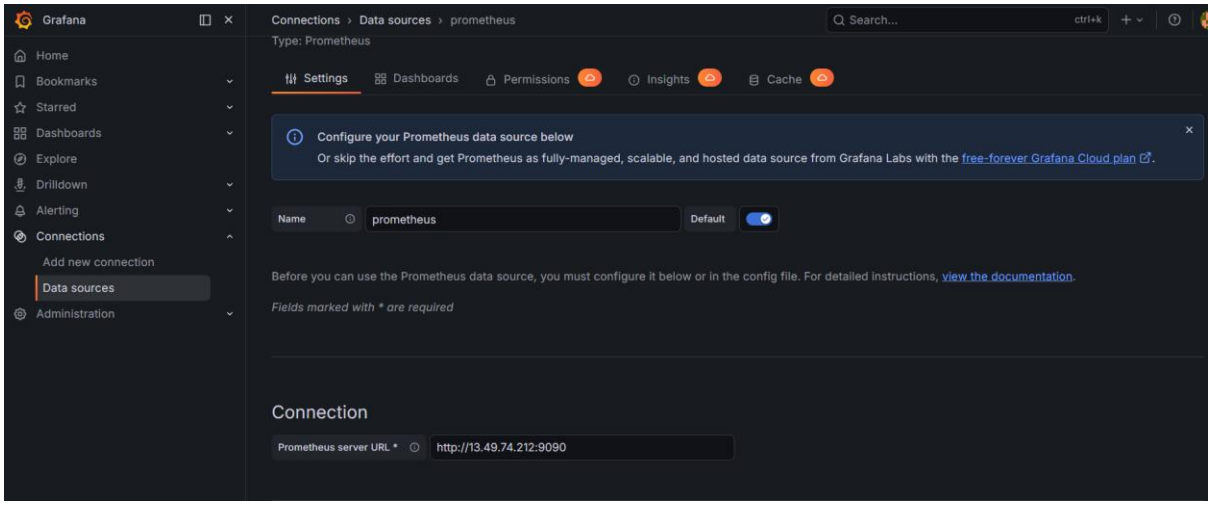
```
info: Adding system user `grafana' (UID 111) ...
info: Adding new user `grafana' (UID 111) with group `grafana' ...
info: Not creating home directory `/usr/share/grafana'.
### NOT starting on installation, please execute the following statements to configure grafana to start automatically using systemd
 sudo /bin/systemctl daemon-reload
 sudo /bin/systemctl enable grafana-server
### You can start grafana-server by executing
 sudo /bin/systemctl start grafana-server
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-24-230:~$ sudo systemctl start grafana-server
sudo systemctl enable grafana-server
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.
ubuntu@ip-172-31-24-230:~$
```

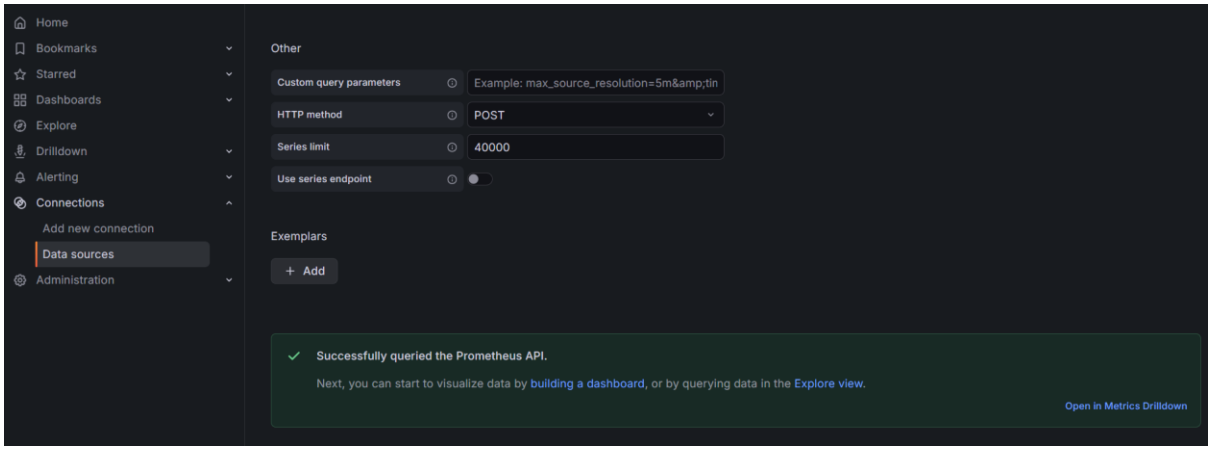
4. Open in browser



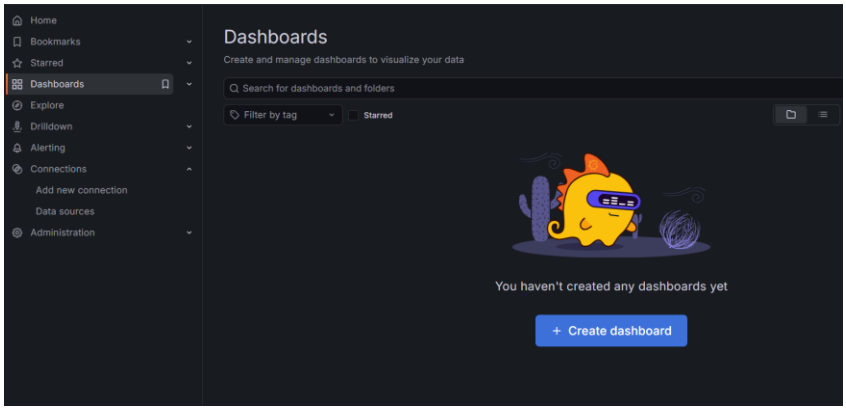
5. Connect Prometheus to Grafana



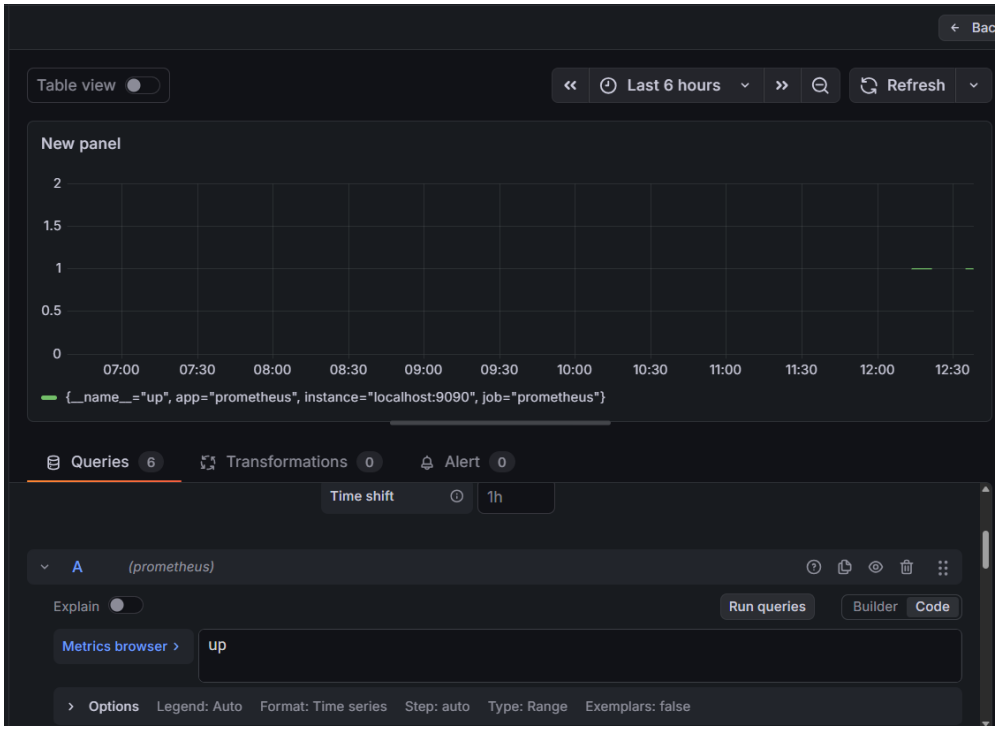
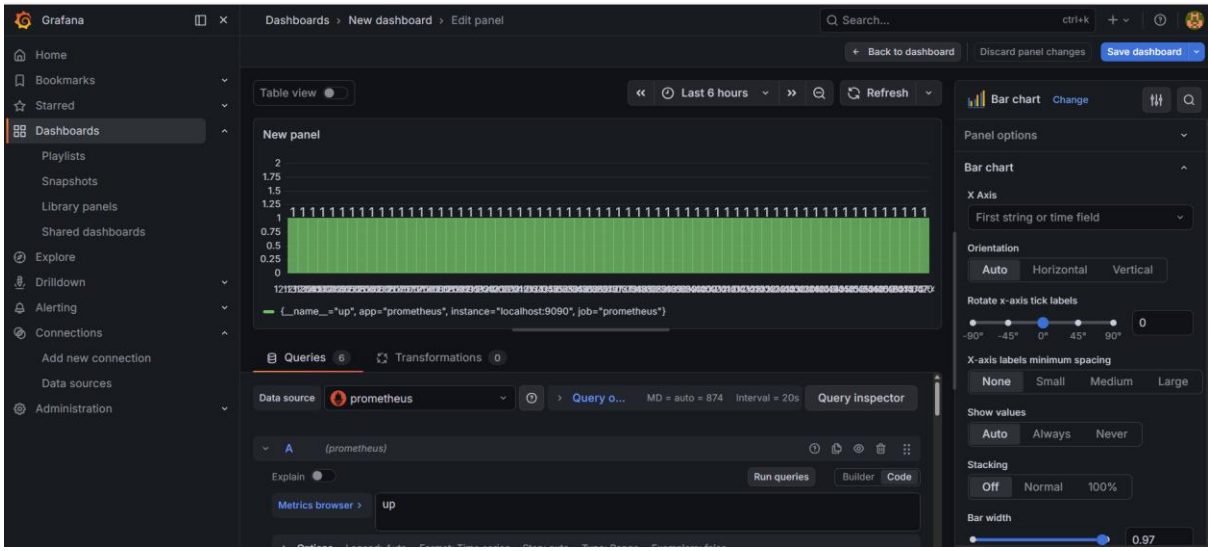




6. Create Dashboard



7. Run query as up



8. Grafana Dashboard